Hints: Abstract algebra – 4. Group actions, homomorphisms and automorphisms

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Orbit–stabiliser: $gx = hx \iff g^{-1}h \in G_x$, so $gG_x \mapsto gx$ is a well-defined bijection. Burnside: count the pairs (g, x) with $gx = x$ once by g and once by x, and use $|G_x| = |G|/|Gx|$.

Hint 2. Necklaces: $G = \mathbb{Z}/n$ acts by rotation on the k^n colourings. A rotation by j splits the beads into $\gcd(j, n)$ cycles, so it fixes $k^{\gcd(j,n)}$ colourings. Exactly $\varphi(d)$ values of j give $\gcd(j, n) = n/d$.

Hint 3. So the number of orbits is $\frac{1}{n}\sum_{d|n} \varphi(d)k^{n/d}$. It is an integer because it counts something.

Problem 2

Hint 1. Cayley: $g \mapsto (x \mapsto gx)$ is an injective homomorphism $G \to \mathrm{Sym}(G)$. Cosets: G acts on G/H; the kernel is $\bigcap_g gHg^{-1}$, the largest normal subgroup inside H.

Hint 2. If $[G : H] = p$ is the smallest prime divisor of $|G|$, then G/K embeds in S_p, so $[G : K]$ divides $\gcd(|G|, p!) = p$. As $K \subseteq H$, this forces $K = H$.

Hint 3. For $|G| = 2m$ with m odd: by Cauchy there is an involution t. Left translation by t has no fixed points, so it is a product of m transpositions, an odd permutation. Hence $\mathrm{sgn} \circ \lambda : G \to \{\pm 1\}$ is onto, and its kernel has index 2.

Problem 3

Hint 1. Yes. Elementary abelian groups have huge automorphism groups: $|\operatorname{Aut}(\mathbb{Z}/2)^3| = |\mathrm{GL}_3(\mathbb{F}_2)| = 168$. Look for a group of order 16 with few automorphisms that contains $(\mathbb{Z}/2)^3$ as a subgroup of index 2 (automatically normal).

Hint 2. Take $G = D_4 \times \mathbb{Z}/2$, where D_4 is the dihedral group of order 8. It contains $H = \{e, r^2, s, r^2s\} \times \mathbb{Z}/2 \cong (\mathbb{Z}/2)^3$. Bound $|\operatorname{Aut} G|$: an automorphism is determined by the images of r, s, z. r must go to one of the 4 elements of order 4; s to one of the 8 non-central involutions; and z to a central involution different from r^2 (the common square of all elements of order 4), which leaves 2 choices. So $|\operatorname{Aut} G| \leqslant 4 \cdot 8 \cdot 2 = 64 < 168$.

Problem 4

Hint 1. Compare the triples $(x, y, (xy)^{-1})$ and $(e, xy, (xy)^{-1})$: this gives $\varphi(x)\varphi(y) = \varphi(e)\varphi(xy)$.

Hint 2. Compare (x, e, x^{-1}) and (e, x, x^{-1}): so $\varphi(e)$ commutes with every $\varphi(x)$. Take $a = \varphi(e)^{-1}$ and verify that $\psi(x)\psi(y) = \psi(xy)$.

Problem 5

Hint 1. Let H_2 act on the n left cosets of H_1. The orbit of the coset H_1 has size $[H_2 : H_1 \cap H_2] = \frac{[G:H_1\cap H_2]}{[G:H_2]} = n - 1$.

Hint 2. So exactly one coset gH_1 is fixed by H_2. Then $H_2 \subseteq gH_1g^{-1}$, and both have index n, so they are equal.